ActInf ModelStream 020.1: Kenric Nelson: "Learning extreme models with the Coupled Free Energy"

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Hello, welcome everyone. This is active inference model stream number 20.1 on January 30th, 2026 and we will be discussing learning extreme models with a coupled free energy by Kendrick Nelson here and colleagues. So Kendrick thanks a lot for joining. you will provide a presentation and if anyone is watching live feel free to put any questions in the live chat and I will read them when we have a discussion. So thanks again for joining and to you for the presentation. >> Thank you Daniel really appreciate you hosting this talk. Uh folk has been doing a deep dive into active inference including reading the the new book that's come out in the last few years. And so we're looking forward to talking with your community about our own work in modeling ex um extremes and how that can be used to improve artificial intelligence. Uh my colleagues that worked on this presentation with me are Igor Alivea from Brazil and Amina Al- Najafi from Iraq. Um and uh and so um I'm what I'm going to go through is sort of an introduction to complex systems and why they have you properties that are unique in terms of trying to do modeling in AI. Um and then I'm going to go through um a variety of entropy measures including the standard ones of the Shannon entropy but also generalizations that apply to complex systems and then show why there's a unique one uh that I call the coupled entropy and how we can use that in variational inference to generalize the free energy function and thereby be able to model quite extreme um environ irements and their distributions and we'll go over some machine learning about um results for that and then I look forward to discussing it with you and uh the community. So fauk is um building capabilities and riskaware intelligence. Uh there's a couple aspects of that. There's a technical aspect which I'm going to brief today with regard to our work in machine learning and risk risk. Um there's also a DAO governance piece in terms of using ideas from complex systems regarding how you uh design protocols for communities to make good decisions. And then we see these two threads coming together in terms of how we can better mitigate um extreme risks like uh the issues around climate change. Uh the team is diverse. I'm I'm in the Cambridge, Massachusetts area. Uh but we have um working partners around the globe and then we also collaborate with universities. Um I was on the faculty at Boston University collaborating with Mark Khan for several years and Eugene Stanley and then I have also collaborated with Sabir Yummer at the University of New Haven and then we have several other uh university partners. I was in Turkey uh for two months this year collaborating with Uger Turknali which was a fantastic opportunity. So here's the challenge um when when it comes to catastrophic risk we know that it's that um those very low probability events that have high impact that are a real concern. But machine learning has grown out of traditions in statistics that revolve around what's called the exponential family and the Gaussian being the most popular um example of that. And even the metrics based are based on the inverse of that which is the logarithmic function that gives us the maximum likelihood metric that gives us the Boltzman gives Shannon entropy. But these distributions and these metrics don't really account for uh the heavy tailed properties and the nonlinear properties that we experience in complex systems. and they're quite difficult because the moments can diverge. Uh so there can be situations where you can't measure the variance um which is the most common thing that you're trying to characterize in in machine learning. So we need ways um to account for this. And just just as one example to visualize this um you know the cryptocurrency space um is one of the areas that we work in as a community and Bitcoin is notoriously very volatile and you can see that in these spikes that are not modeled by a Gaussian distribution. Um here we have in the middle an agent-based model where in red you would see something like a Gaussian process and you can see that yes there's noise but there's not these large fluctuations but we can model u these with a distribution that I refer to as the coupled Gaussian and the coupling has to do with the fact that I we can map the tail decay shape to the source of nonlinearity ity in the system and I'm going to go into more detail about that. Um so [snorts] we're generalizing the exponential family and just um to remind people what that is. First of all of course it evolves around the exponential function and then you have a set of parameters that are dot multiplied with a set of statistics. the traditional ones being just x and x squared. There can be some other terms that involve x uh but we'll typically just set that h to one for today's discussion. And then all of this is normalized by what's called the partition function. When we go to the coupled exponential family, what we're doing is we're modifying this exponential function to be a power law type of function. And I show that here. So this coupled exponential, I'll highlight it right here. Oops, I got ahead of myself. Um, [clears throat] it multiplies the variable by this coupling, adds one, and then renormalizes or rather takes it to a power one over the coupling. And that plus one is a critical feature of the math. That's what enables it to converge back to the exponential. But it's also what complicates things in terms of analysis of nonlinear systems. Now we keep the statistic and the parameters. Um and there's an additional factor here. This 1 + d kappa has to do with the the change from a cumulative distribution. When you take the derivative uh you get this additional parameter in the exponent. The most common distributions that we'll talk about today are the generalized paro distribution which I refer to as the coupled exponential. That's when this alpha term is equal to one and the student t distribution when alpha is equal to two and I refer to that as a coupled gaussian. The critical thing is that the moments diverge. Um and this as this coupling grows more and more of the moments diverge. And I'll I'll go into more detail of that as we go forward. But um taking a look at an example here that's going to be used a lot in machine learning because this is the distribution that will you know be our latent distribution is the coupled Gaussian. So in black here you'll see the traditional bell curve with the coupling equal to zero. That's your Gaussian and that's what traditional machine learning algorithms are built around. In blue is what's called a compact support distribution where it actually decays faster and truncates. I'm not going to talk a lot about that today, but that would be a negative uh coupling. And then what we're really focused on is these heavytail distributions. In this case, the one in red has a coupling of one, which is the Koshy distribution. And the critical insight in defining this family is that the shape that tail shape parameter is exactly equal to the nonlinear statistical coupling or the source of nonlinearity in the system and these are proportional to a measure of complexity of the system. So we're going to show that we can model now the most extreme complex uh systems with these techniques. One of the early insights about this work that I showed in a paper several years ago um that actually introduced the coupled entropy function was a concept of an average density. So it's people are fairly familiar that the center of a distribution is the mean and that the variation is expressed for the Gaussian as the standard deviation and more generally for these non-exponential distributions it's referred to as a scale. But people have a difficult time uh conceptualizing what entropy is measuring. You should think of entropy as the measurement of uncertainty along the yaxis of a distribution. And you can always take the entropy which is in the log space, take the inverse of that, the exponential, and bring it back to the probability space. When you do that, you you [clears throat] get a function that I refer to as the average density of the distribution. And this is a translation of a generalized entropy function. And what the key thing here is that that average density always lines up at the scale. This is the invariant. And so this particular scale of the distribution has a number of things that are unique about it. Um, and in particular, what

I've proven this past year is that while there's literally dozens of candidates for generalizing the Boltzman Gibbs Shannon entropy, there has during this 50-year period of development in the modeling of complex systems been a missing requirement. And that requirement is that your generalized entropy has to be a measure of this particular point of the distribution where you have what I call the information scale. And the reason that's important is because that point in the distribution is sensitive to the linear sources of uncertainty whereas the tail decay is sensitive to the nonlinear sources of entropy of uncertainty. And these two parameters are the only way to precisely separate those the role of linear and nonlinear processes. So let's take a closer look. This is these distributions here are from the family of the coupled exponentials or the generalized paro distribution. So if we start at the top, this is one of those compact support distributions and we can see it for two different um parameters of the scale. On the top the scale is equal to two and on the bottom the scale is equal to half. And you can see here clearly how it truncates. Um now [clears throat] the Boltsman gives Shannon is in that case lower than this scale point and when and when we go to heavy tail these lower curves what happens is the Boltzman gives gets its probability gets very low and that is to say its entropy gets very high and it becomes dominated by the tail shape and you lose sensitivity to this scale parameter. So this is what motivates the need for a generalization. And the first um significant generalization was done in the 1950s and60s by Alfred Renee where he showed that you could match a generalized mean to that tail shape and then still take the natural logarithm and get a very effective measurement of uncertainty in these complex systems. And indeed it's a good metric. it's it's used quite a lot um in machine learning and in other applications but it doesn't match up with this what I'm saying is a requirement here to actually measure the the shape the the uncertainty right at this scale there is another significant proposal that dates back to 1988 and has gotten a lot of attention recently known as the solace entropy um it introduces a concept that I'm going to talk about more that's quite important called an independent equals However, I've since proven that it is not in fact the correct function for generalizing entropy. Um, and there's a missing piece to it that was left out. But if you look at this requirement, um, this is what I'll show is equal to this coupled entropy that I've defined. I'm going to show that in a moment. But the but the fascinating thing about this is that the solution right along this scale is equal to 1 plus a generalized logarithm of the scale. And and this is the beauty of it and this was sort of the Eureka moment when I saw this solution that I realized that there is something quite unique about the coupled entropy and about this requirement because now I'm maintaining in my entropy function a a a dominant metric about this scale parameter and the only thing I'm doing is I'm changing what logarithm function is measuring that. So let's dig into it a little bit more and see what is the where this function comes from. So as I said we start with a coupled exponential function that forms these non-exponential family of distributions and then we take the inverse of that to form the metrics uh known as the coupled logarithm. And here you can see clearly that the coupling is directly that power and there's a minus one in order for it to converge properly and there's a normalization. Now when we go to forming the entropy function there's some additional features that have to come into this. So the first thing you'll see is that this generalized logarithm now is the coupling times alpha. You can think of alpha as the shape near the location of the distribution. So again the most common ones is one for the exponential distributions, two for the Gaussians, but it can be a real number. The other critical piece and this is what was left out of the formation of the solace entropy is that [snorts] you can't just take this natur this generalized logarithm of the probabilities. You have to account for how dimensions affect uh the metric. And this again has to do with the fact that you can think of the coupled logarithm and the coupled exponential being appropriate with the cumulative distributions. But once you take a derivative uh you have to account for additional terms in the exponent and each d each dimension involves another derivative. So you get this factor here that you have to raise the probabilities to. Now this term in the right this generalized logarithm um let me go back that is exactly equal to the inverse of this um generalized exponential and so what that gives you is the interior of this and you can bring this inside the parenthesis too I'm not going to show that today but basically you're going to get um the argument of this distribution however However, that's not enough. There's another piece here that pulls the whole thing together, and that is that you do not average over the distribution itself. you average over something that's uh I refer to as the independent equals uh distribution because what this is modeling is how many random variables are um sort of independent but equal to each other and this is the metric that will give you the parameters of that distribution. So this is the this is the proper statistic that function I can show here. It's it's taking the original distribution P and raising it to this power and then renormalizing. And the effect of this particularly all the way here on the right is that it reduces the original coupling value. So now I have a modified coupling value and the ca the critical thing is that as you go to infinity you always you know for for alpha equal to two is the case we're going to be most interested in this will converge to a half and that half is the boundary where the distribution would go to infinite variance. So, so now what I have is this balanced metric in which I'm going to put a very high penalty on low probability events through this generalized logarithm.

But at the same time, I'm going to sample over a distribution that makes sure that those rare events don't occur too often. So let's see what the impact of this is now on variational inference. Um if you know Dan if there's been any questions in the chat this would be a good place to break and and maybe answer some questions. Um >> let me just ask one clarifying question. What is coupled what is coupled to what? >> Yeah. Okay. Excellent question. So I originally coined this phrase nonlinear statistical coupling because you can think of it as treating the states of a distribution as not being independent from each other but rather the states are coupled together. Um it comes out explicitly from this um coupled exponential dis this independent equals distribution. You can kind of rearrange this function and show that it's also equal to multiplying all of the states together where you have the state you're interested in but the other states are raised to this power. Now since then what's become more uh relevant is the fact that this coupling term is the source of nonlinearity in the system. So a classic example which I'm not going to show today but is very relevant to the fluctuations in Bitcoin is multiplicative noise. So um our our line when we study linear systems we typically have a deterministic component that we're trying to learn and then we model say the Gaussian noise as being additive. But in complex systems you're going to have a third term in which the deterministic process and the noise process are multiplied together. and you can show that um the coefficient on that third term is equal to this coupling. >> Okay, so just to kind of echo that back in a Gaussian mixture model, whether you have just one Gaussian or multiple Gaussians, you're superimposing different means and variances, but the variances never interact. It's just kind of like you're stacking a bunch of Gaussians on top of each other. Whereas here we can get something that's looking more like multiplicative noise because of this multip um coupling that that extends the tails of the distributions and >> yes >> and I guess maybe address it at the end or when right but how do we know how much to inflate those tales when how do we calibrate that factor to empirical ical data and how would that differ in its calibration or or precision or recall than just using one of these other traditional heavy tail distributions. >> Yeah. All right. So I that's a good I'll answer that now as a preview of where we're going to go with the talk. Um so one thing we haven't shown all of this but I would argue that this coupled exponential family has the ability to model any heavy tailed distribution because what I'm doing is that coupling parameter is um the asmtoic tail decay. So you can figure out what that is and then the parameters and statistics measure are measurements of all all that's happening in the distribution in between. So there's not a limitation in what kinds of distributions you can model with this method. That's number one. Number two is a very tricky question. Um, I'm going to talk about the fact that the way I see it, there's kind of two purposes for the coupling, two or three. Um, the first, as you're describing, is the system itself that you're trying to learn can have these fluctuations, and you're going to want to characterize that and then use that. In the way we're doing it right now, it's a hyper parameter, but you can you can analyze the data and and determine a tail that fits to the data you're interested in. That's one. But I think actually the more important thing is a way to make systems robust because whenever we train, we always have limited data and we also also have data that might not exactly fit what we know we're going to experience out in the field. So uh we can also use this tail decay as a way to improve the robustness against things that weren't in the training set. So there's these two components to how you will use the tail. All right, with that I'll press forward. Um, so we're particularly interested in how we can apply this to variational inference and active inference. So variational inference is a basian learning method where we try to approximate a complex true posterior distribution with a a tractable distribution where we're going to uh learn the parameters of that distribution and the goal is to minimize the divergence between the true posterior and the and the approximate model. Now recently and what Daniel is uh and his team at the active inference institute is focused on is that this basic framework has now been applied much more broadly to neuroscience, artificial intelligence and even uh socioeconomic problems where you can think of every single node in a neural network as doing a comparison between a top- down prediction or a bottomup observation and then the difference becomes a way to um build an inference model as you optimize and minimize those differences. Likewise, you can apply this to actions and policies by using the prior or the prior top down um as a preference of what you want to have happen and then the observations are now expected observations and then that can be that can control uh the actions of a system. So the challenge is how can we modify the cost function this EV evidence lower bound or the negative free energy uh to be able to learn heavy tailed Um so let's we're in our own experiments to date we've been using the variational autoenccoder as a foundation for the experiments. So let me just go over what that is. Um here you can take a complex data set such as uh the celeb a uh photographs and you use a neural network to encode a mean and a coariance matrix. Again, in the standard VAE, this is a Gaussian or or a mixture of Gaussians that you're modeling in the latent space. And then you can sample from that in or decode it in order to get a new sample generated. And the thing that you're optimizing here is what's called the negative elbow or the free energy. It has two components. It measures the quality of the reconstruction and it measures how simple um your latent space was in terms of staying constrained to traditionally a mean vector of zero and a coariance of one. Now uh we can generalize this and what we do now is we're going to use a coupled Gaussian and then we're going to match that coupling with a metric which is the coupled free energy and it turns out while the derivations are complicated in the end we get out the same metrics that are used traditionally with some additional coefficients. So if we assume the pixels are continuous approximately then um uh well let me start actually the latent layer then is measuring the divergence and that's basically just the trace over the um co the the what's now called the correlation metrics and then also the differences in the means between the prior distribution and the posterior distribution. So that's one component and that is a just a calculation and then uh we also measure the mean square average and again this isn't changed it's just some coefficients that have changed it's it turns out we still get the same metric but now what happens is because we have this heavy tailed distribution as the model um we're this mean squared average is going to be highly sensitive to what's happening in the tails. And the way we balance that now is that rather than sampling the latent space directly, we rather sample this independent equals modification to it. And this is the real innovation that's made a big difference. Now um and again what happens is as the coupling of the Gaussian goes to infinity the coupling of the samples we're going to generate is limited to less to be less than 1/2. So we always guarantee that there's a well- definfined mean and variance uh from the samples that we're uh drawing from. So let's see how this works. show a couple of pictures of this. So on the left I have um a log linear plot and then on the right is the log log plots. And if we start with this light blue curve that looks like a parabola here this is the Gaussian and the coupling is 10 the minus 5 which is essentially zero and the modification doesn't really change that at all. It's still just 10 the minus 5. And again, we see this Gaussian here as a knee on the log curve and a very sharp decay of the exponential. Then if we go to a Koshy distribution in this dark darker blue, we see this very flat tail curve here and you see the longer tail here on the log log plot, but the knee is still in the same place. Now when we take that independent equals curve um we have a much softer decay or excuse me we have a much faster decay here. Now the coupling is 1/3 and again that's less than a half which is the critical point here. So we took a distribution the Koshy that h doesn't have a defined mean the variance is infinite and we modified it into a distribution that's well behaved and again so we see that as this sharper curve here on the log log plot and then finally we can go all the way to the extreme of a distribution with 10 to the five coupling. There's not necessarily practical use cases for s such an extreme distribution but the point that we can actually still have convergent learning with such an extreme distribution is a unique capability. So we start with this sharp spike here, this very long tail. But both the coupling and the um correlation matrix get modified such that the modification is more like a delta function here showed here in red with the dash. And you can see that on the right here the very sharp turn here has to do with the fact that the coupling is so high. But now um we have a curve that's more balanced here at the top. And again this uh new curve or distribution has approximately 1/2 as its coupling value. So what does that enable us to do? Um going back to this example of modeling the Bitcoin returns in traditional machine learning with the uh CVAE we would be using a Gaussian as the model and it's it's a mixture so it's going to get a little bit of the tail but ultimately it fails to capture these extreme elements of the Bitcoin log returns. Um so the blue is the actual data and then this sort of um orangey gray color is um the model. If we set our coupling to be 6 which we measured to be approximately what that uh particular uh set of data was characterized by. Now we can see that we do have uh samples all the way out into these extreme tails. And so this is the key point here now is that we can we can now have a way to model and learn um these extreme distributions. Now to date we we'll make progress on this but to date the coupling is a hyperparameter um that we're setting a priority to the training and that but what is important is that we can then learn that correlation matrix um to fit uh the data. Is this still overd dispersed on the Bitcoin there? >> The model isn't perfect if that's what you're referring to. Um so yes, it you know um it does look like we're getting um some charact some samples that are further out than what the data was. Is that what you mean? But >> yeah, and and then and then again just how would this differ than than an inflated tail? >> Was it does it have a different computational requirement or does it just move probability density from a central tendency >> out to a tail in a way that has a certain um I guess curtosis. >> Yeah. So let me let me talk about that in two ways. If we stayed with the Gaussian distribution, uh the problem is that we would end up with this very fat distribution potentially like if we really wanted to capture this sample out here, we'd have to have a very fat standard deviation and then that would mischaracterize the fact that most of the samples are in near the mean. Um, if you go, let's see, if I go all the way back to this curve, we often talk about the tail being slower, but another feature of these heavy tail distributions is that around the mean, it's more spiky. So, there's a lot of mass near the mean and then it spreads out. And so what h so so it's both of these characteristics that you want to be able to capture in your model. So that's one aspect. So the Gaussian isn't going to be able to do this. Now there's a second question. Oh, is the the second point is the following and maybe the easiest graph to show this is if I go to this graph here the question might arise why can't I just use a heavy tail distribution in the latent space but keep the Shannon entropy metric right well the problem is what happens is that the Shannon metric is going to be so sensitive to uh the uncertainty caused by the tail that it will not have any un it will not be sensitive to the scale which is what you're trying to learn and modify the parameters of in your neural network. Um and in particular what happens even when so I'll go one other thing here in the early work we have a paper we're we're writing a new paper on this currently there's a paper that we published around 2020 on the coupled VAE where we used the generalized logarithm but we sampled from the latent distribution we didn't really use this part of the function And what we found there is we were limited to around a coupling value of 0.5 to one. And as you tried to go to higher values of coupling, the training process just wouldn't converge. Uh so eventually you'd get um such extreme measurements that the training process would just crash and and you wouldn't have any learning process at all. And so to date, the state-of-the-art is that yes, you can train heavy tail distributions, but typically they're limited to distributions that are of uh faster decay than the Koshy distribution, you know, which is coupling equal to one. And it really doesn't work very reliably unless the coupling is less than 0.5. And I I'll just jump ahead here. What we're showing now is that even in this most extreme case of 10 to the fifth where you can I'm not convinced that it has practical value but it still converges. I still get a a neural network with a model. Um so this is showing now um let me let me just back up a second. So we're we're doing this test over the um celeb a data set uh where we have hundreds of thousands of images of celebrities. Um and again the key thing here is that in this latent distribute I'm going to learn the latent distribution and after the training I am going to sample from that latent distribution directly but during the training process I'm going to sample from this modified distribution to guarantee that the training process converges. Um and we show that in fact it does. Um so against this metric of the coupled free energy we get continual improvement as we um increase the coupling all the way up to 10^ the 5 and at 10 to the 5 it's just a flat line here. Now there is a caveat that um this is not a onetoone comparison because we're modifying the model and we're modifying the metric. Um but the important thing is is we do show that you know the amount of variation we get as we as we increase this coupling value gets quite low and and also just it's I think it's a brand new result just to be able to learn anything about a distribution so extreme >> what is the the free energy there the yaxis >> the yaxis that is this coupled free energy energy. So it's the generalization of the free energy. And again, the way it is is that as we go to higher coupling values, it's more sensitive to what's happening in the tail of the distribution. So you can think of it as like a robustness metric, but at the same time, we've modified what what samples we're drawing over in order to make sure that this very stringent metric can still converge. Um now in order to do uh we we have then looked at metrics where we keep the metric constant in order to do the um head-to-head comparison. And here we see what we would kind of expect is that we get improvements in the accuracy. And I'll call coupling equal to zero still a accurate measurement. Whereas the positive values of coupling I'll refer to as robust metrics. We do get um as we go from 10 the minus5 to 10 the minus2 we do get improvements in the accuracy and throughout the whole epic um we're getting an accurate model. Um and that is up to I guess the I think the darker one here is um 10 to the 0. Oh actually say 10 to the minus one is the darker one. Now when we go to the Koshy distribution and beyond the Koshy distribution we do see that the accuracy metric with this free energy does show decays in the performance. So you in this case the training has kind of a optimized point very early in the epic cycles and then as you continue to train it the model actually gets worse with regard to accuracy. Now that's not unexpected because we weren't optimizing for the accuracy. We were optimizing for something else. Um and then the other thing this is labeled incorrectly but the um the last one here is the extreme case of 10 to the five and it does show that this extreme model has kind of funky behavior and that that's sort of typical of what would have been expected but again I emphasize without that use of the independent equals the whole thing breaks down it more like 10 to the uh 10 to the 0ero here that was sort of the limit of the prior results. Now we also looked at how uh the reconstruction fidelity is for this um and the on the left is the original samples in the middle is the uh traditional VAE with the coupling equal to zero and by eye you can maybe make out a little bit of improvement with the coupling value but the metrics are actually quite impressive. Um so against a metric known as the multiscale structural similarity um we see 25% improvement and that stays uh consistent across all of the coupling values. So even just a little bit of coupling getting away from that the extremes sharpness of the exponential tail right away you get performance improvement and that performance improvement stays consistent across all the coupling values that we used. Um likewise um something called the learn perceptual image patch similarity had a 10% improvement. Um and I guess okay so that that's it. So the conclusion here is that we now have the ability to model the most extreme distributions and we're particularly um interested in how this could be applied to catastrophic risk modeling number one. uh but also I I have a lot of scientific interest as in terms of how it will improve active inference for the following reason. If you're going to measure errors in every single node. So again just remember the contrast here. Right now the way machine learning algorithms are trained is that you have one cost function that's going to um be used to modify the parameters of a large network. Um you can argue that in that situation uh the Gaussian and the entropy measurements may be adequate for most use cases. Maybe not one if you were trying to model catastrophic risk, but for other use cases, we've seen that it's very powerful technique. This is the engine of AI. But in active inference, because you're measuring errors at every single node, you necessarily have very limited amount of data characterizing those errors. And that's exactly the original use case for the student t distribution was that a low amount of data corresponds to a low degree of freedom. And it's and that degree of freedom is actually just the inverse of what I'm calling the nonlinear statistical coupling. So another way that these models are grounded in traditional statistics is that uh the nonlinear coupling is just the inverse of the So I think that in the active inference models um there's a real necessity for this type of metric because there's a necessity um to be able to uh model accurately the fact that you're trying to take these error measurements over a limited amount of data. and and the novelty here is to have both a generalization of the metric in this coupled logarithm and a generalization of the samples you're using with this independent equals um and new work that we're making progress on is uh the theoretical proofs um using information geome geometric techniques you know that the this is exactly the correct statistic Um, and I'll also say like just for maybe going to future reading here. So, I have a preprint that's currently under review that provides the proofs that there's a unique scale called the independent uh excuse me um that I refer to as theformational scale that is required for the modeling of heavy tail distributions. And then from that there's only one generalization of the entropy function that gives you the correct measurements. Um and so the paper is called on the uniqueness of the coupled entropy. And this a variation of this presentation was presented with a paper at the um artificial general intelligence conference 2025. Um we have a new paper that's will have more details coming out soon. Uh but the original paper um from a few years ago is called coupled vae and there we showed improvements in the robustness using these techniques. And then another interesting paper I wrote several years ago was a a book chapter um called reduced perplexity which was a play on words um but it was showing this idea that you can think of entropy as your measurement of uncertainty along the y ais of a So with that I can uh thank you all. Uh really appreciate the opportunity to talk to this community, learn more about what you're doing and looking forward to some nice collaborations together. >> Great. >> Yeah. So people watching live can write some questions. I'll ask some some random things. So these are all central tendency distributions. How do you deal with situations where you just are not mean centered around zero? >> Yeah. Okay. Excellent question. Um, so alpha equal to one is the generalized paro distribution which typically is a one-sided distribution. Um, so that could also be the model. Um and then um with Uger Turkelli we are doing um some research now in which alpha is not equal to an integer and those models have different kind of characteristics. That's number one. The other thing is that the family itself includes that um H of X parameter. So another generalization is what we call the coupled wel distribution. So the wyel distribution is another popular one-sided uh distribution from the exponential family and then you can generalize that with this coupled exponential. So we're also looking at that as an interesting case. Yeah. Okay. Okay. Now, >> I have um I'm happy to take questions from the audience, but I have a question for you, Dan. In your work on active inference, have you noticed this feature that because you're trying to learn at every node of a network that you have difficulties with the limited amount of data that each node um experiences. there's nothing about these models that intrinsically has low or high amounts of data. You could train a massive data set on a single parameter or or have a situation where there's very little data. um not always just a low number of data points per parameter to learn like the sort of large uh p small N issue, but especially getting into thinking about causal effects of action, >> having world models or just having generative models that have a lot of conditional aspects makes it such that you will not have data for a large number of conditional aspects which sort of forces factorizing or simplifying the problem back down to simpler uh lower dimensional subpros. M >> and so the challenge would be well if those lower dimensional subpros could be solved but then you have even a lot of combinotaurics of which smaller dimensional subpros to tackle and again if you knew the structure of the problem then that is very tractable structure learning. But then in a situation where you might be dealing um with just different kinds of data streams and not necessarily have a well ststructured world model then even that structure learning problems becomes very challenging. >> Yeah. Uh well and yeah you actually bring up another interesting use case which is the curse of dimensionality. So as you go to higher dimensions, this issue of needing to use the heavy tails to in order to make sure that your models are more robust and more accurate becomes more prominent. >> On the heavy tails, I wondered, have you thought about any of the physical connections with the entropy? So there's obviously more generalized andformational entropies than necessarily happening in a cup of water. But how do these different tail distributions, for example, connect with temperature distributions and thermal movements of particles? Do they all follow one type of tail distribution or do different materials or different physical situations follow one of these differently? >> Yeah, this is actually the heart of what scientists in the field of complex systems work very actively on and you know thermodynamics grew out of modeling equilibrium u type things. So, so temperature for instance is a measurement that re basically has a requirement that you're in equilibrium. But of course, we're very interested in being able to characterize nonequ equilibrium systems um or what can be called nonequilibrium steadystate systems where it's not in equilibrium but it does but the state does persist over time. Um and there's a lot of work in that area. Um this is a big part of what motivated a field called non-extensive statistical mechanics that was originated by Constantino Solace and there's been proposals to generalize thermodynamics using modified entropy functions. Now I would argue but it's not fully proved that mostly those attempts have not been very successful. They've had modest success, but they've not really come up with a complete model. And in this new paper, I have a section showing that you do get a you get a rigorous definition of the temperature as being equal to this scale parameter. Um and again this is another thing that I think makes both the way I'm characterizing the distributions and this particular entropy function unique it because I show that you can um that basically that scale parameter is your generalized temperature. Um and the the interesting thing about that calculation is that uh in order to get that what came out of the mathematics is that the free energy function um that underlies thermodynamics it's that function that gets generalized and has an additional nonlinear term in it. Um and so that's where the generalization comes and then the other thing is I showed that there can be a complimentary model where I can define the temperature to be uh the scale raised to a power related to that coupling term. Now my definition of temperature is more complex and involves the nonlinearity. But in that case then I can um have no modifications to the free energy equation. So there's also sort of this interesting duality in terms of how you build the models. But I think with that result there's a strong case and it would be very worthwhile for people to look at whether the full laws of thermodynamics can be generalized uh using the coupled entropy. back to to the data, it's interesting that you highlight learning from little data whereas almost one of the key advantages is that you will not deflate the variance when you have a lot of data because the main issue is not having a wide estimator on sparse data. That's free. Mhm. >> The hard thing is when you have a lot of samples not to just converge back to a training data set. And that's why variational methods and all other kinds of entropy regularized methods kind of are out there because you don't want a collapse simply to recapitulating the training data or even narrower. >> Yeah. Yeah. Well, I Yeah. You know it was it's funny how just in the process of working on these things you learn a different connection. So I have my background is in electrical engineering and as part of that I've studied both mathematics and physics quite a bit. So I had not I would say a strong background but certainly a background in thermodynamic processes. At the same time as we were um developing the variational autoenccoder we were learning about the importance of the evidence lower bound and it's uh basic way that it can be decomposed into a measure of accuracy in your generative process and then also a measure of the regularization of making sure that you using a simple model and that it's the balance between these two things that's important and ultimately what gives you a good model. So it's not just accuracy but it's accuracy with the simplest model possible. Now it was only through um it was only through reading Carl Friston's work on the active inference that I had this got gained this understanding that the evidence lower bound function is exactly the free energy. And so there is a direct connection between the thermodynamics of complex systems and the training of artificial general intelligence and generative models. And so I as we as a team have been reading um Parr and Kristen's new book on active inference, it's been fascinating to see just how general the concept is. And you know again another thing that's exciting for us is that we've mostly focused on variational inference. And I six months ago I was sort of expecting that as we migrated to decision making and you know kind of robotic control that that would require a whole another set of algorithms and a whole another set of generalizations that we would have to work on. But now of course active inference has integrated all of these techniques. um you know so so now I I have a better understanding that um when it comes to action you can think of your utility function as just a prior distribution and then what action you want to take you can think of as an expected um minimization of the free energy. So, it's a beautiful representation and and we're really excited about working on these different kinds of problems. >> Let's sort of foreshadow and head off on one area that we've talked a little bit about, which is when we're dealing with causal models and structure learning, we're in a little bit more of an expanded setting than just like a financial returns model. So in a financial returns, we're dealing with something that might be tightly clustered or widely dispersed or have an average trend or be zero centered on a given fixed, you know, 24 hours percentage change. >> But and and as you get there's more and more ways to be extreme and there's fewer and fewer ways to be essentially conformist. Now when we open the door to cognitive systems and multi-agent systems and we're thinking about complex systems in general. So there's more and more and more and more ways that couplings amongst actual subunits can interact. So what what kinds of macro properties or approaches might help us bound where we have uncertainty not just over curtosis properties within a distribution but where we actually have a structural uncertainty about like geospatial causal models. I'll answer this a little bit indirectly but I think something fascinating that comes out of this is another core principle in the active inference um model is the idea of a marov blanket where you separate your internal model from the external model and then a market blanket is where you declare independence and and sort of And and another way it can also be thought of like one way you can think of it as the markoff blanket between the past, the now and the future, right? Um well that's certainly foundational, but one of the things that comes out of this coupled algebra is the ability to relax that assumption of independence. Because if I apply now rather than a markoff blanket, what I might refer to as a coupled markoff blanket, I am going to have independence in terms of those correlation matrixes. um but I'm going to have be able to maintain a model in which there's actually coupling uh across the blanket that so so the coupling parameter is now going to have be a way of storing long-term correlation um and I have done a little bit of work on this in papers that either refer to coupled markoff processes or coupled basian processes and and and at the heart of each of these again the basian calculation is similar. One of the basic assumptions in basian analysis is that the prior distribution gets multiplied by the likelihood and then that gives you the posterior distribution. Well, that multiplication is assuming that the prior and the likelihood are independent of each other. But with the coupled algebra, you can relax that and you um you can have a a nonlinear coupling between your prior state and your um posterior state. So for instance um there's a scientist Topsoy who's used that model in models of cognition and and the way I think about his models is sort of like how how strongly you care about your prior. And so you you know you you you might be a person that's sort of tied to a particular set of beliefs, right? And no amount of data is going to change those beliefs. And you can model that with this coupled algebra. And and then vice versa. might be someone who doesn't have um you know you're very loose in your priors and and you're very loose in your beliefs and you kind of then easily swayed by whatever data comes in next and these kind of differentials um are some of the more interesting models that can be created with this mathematics. Now I don't think that answered your question directly but uh it it but the point is that there's much more subtle kinds of causal models that can be created uh using these techniques. >> Well, I like the idea of the marov blanket as the declaration of independence. A fun idea and definitely the attention to past, present, future or just the marovian property of time with the past influencing the future >> through the present with this basian moment of the intersection of how you couple the prior and the likelihood. And then also in in in that it makes me think about yes we could talk about all of the interactions among subunits and there's combinotaurics of structural possibilities for the system of interest. However, the variational modeling agent may have a much simpler model that they would have more defined properties of. So the estimators model properties the map do not need to recapitulate actual structural adjacencies or aspects of the world and intelligence is about that kind of basian information criteria or similar driven approach to ride on that edge with a dual imperative of accurate and small. >> Mhm. Yeah. Ultimately, it is part of the goal is to get simpler models. I I I agree with that. And one one way you can picture that is that if you have these heavytail issues, one way you can get a model still using the Gaussian is to have a mixture of many Gaussians with different uh variances, right? And and so in a sense what the heavy tailed is giving you is the ability to have one distribution that can characterize a a large mixture model of many Gaussian uh distributions. So, so it does, so I do think that as you incorporate the heavy tail into the the foundation of these models, you'll also be able to simplify how how many mixtures you need in order to get a good characterization. Just last comment or note here. Um what what kind of software tooling or what repos or which of the papers should people explore for if they wanted to get hands-on? >> Um right. Um so the two papers on the machine learning um have references to our nonlinear statistical coupling and CVAE uh repositories on git github. Um, and we the the NSC library um just uses a basic uh Python tools and then we originally did the work in Tur um I'm TensorFlow and then we migrated more recently to using PyTorch. >> Cool. So I hope that people >> walk through this and explore the tooling and um >> if there's anything else please feel free to add it otherwise this has been very uh interesting connect some of these basic statistical principles basic but advanced statistical principles to some of the real situations that embedded systems and intelligence systems are finding with uh you know too little too much data events that have more variability than we have seen in training data. These are issues that beset all. So, it's awesome to see how you're researching and applying that. >> Yeah. Super. Thank you very much for the invitation, Daniel. >> Okay. Farewell. Bye. >> Take care.